distinguish the two. A common notation is to use Ω (an upper case omega) to represent frequency in the z-domain, and ω (a lower case omega) for frequency in the s-domain. In this book we will use ω to represent both types of frequency, but look for this in other DSP material.

In the s-plane, the values that lie along the vertical axis are equal to the frequency response of the system. That is, the Laplace transform, evaluated at $\sigma = 0$, is equal to the Fourier transform. In an analogous manner, the frequency response in the z-domain is found along the unit circle. This can be seen by evaluating the z-transform (Eq. 33-1) at $r = 1$, resulting in the equation reducing to the Discrete Time Fourier Transform (DTFT). This places zero frequency (DC) at a value of *one* on the horizontal axis in the s-plane. The spectrum's positive frequencies are positioned in a counter-clockwise pattern from this DC position, occupying the upper semicircle. Likewise the negative frequencies are arranged from the DC position along the clockwise path, forming the lower semicircle. The positive and negative frequencies in the spectrum meet at the common point of $\omega = \pi$ and $\omega = -\pi$. This circular geometry also corresponds to the frequency spectrum of a discrete signal being *periodic*. That is, when the frequency angle is increased beyond π, the same values are encountered as between 0 and π. When you run around in a circle, you see the same scenery over and over.

Analysis of Recursive Systems

As outlined in Chapter 19, a recursive filter is described by a **difference equation**:

EQUATION 33-2
Difference equation. See Chapter 19 for details.

$$y[n] = a_0 x[n] + a_1 x[n-1] + a_2 x[n-2] + \cdots + b_1 y[n-1] + b_2 y[n-2] + b_3 y[n-3] + \cdots$$

where $x[\ ]$ and $y[\ ]$ are the input and output signals, respectively, and the "*a*" and "*b*" terms are the **recursion coefficients**. An obvious use of this equation is to describe how a programmer would implement the filter. An equally important aspect is that it represents a mathematical relationship between the input and output that must be continually satisfied. Just as continuous systems are controlled by *differential* equations, recursive discrete systems operate in accordance with this *difference* equation. From this relationship we can derive the key characteristics of the system: the impulse response, step response, frequency response, pole-zero plot, etc.

We start the analysis by taking the z-transform (Eq. 33-1) of both sides of Eq. 33-2. In other words, we want to see what this controlling relationship looks like in the z-domain. With a fair amount of algebra, we can separate the relation into: $Y[z] / X[z]$, that is, the z-domain representation of the output signal divided by the z-domain representation of the input signal. Just as with